The acceleration is along the negative y direction.

$$\vec{a} = g\left(-\hat{j}\right) = -g\hat{j}$$

Note For convenience, sometimes we take the downward direction as positive Y-axis. As a vertically falling body accelerates downwards, g is taken as positive in this direction. (a = g)

2.10.3 Equations of Uniformly Accelerated Motion by Calculus Method

Consider an object moving in a straight line with uniform or constant acceleration 'a'.

Let u be the velocity of the object at time t = 0, and v be velocity of the body at a later time t.

Velocity - time relation

(i) The acceleration of the body at any instant is given by the first derivative of the velocity with respect to time,

$$a = \frac{dv}{dt} \text{ or } dv = a\,dt$$

Integrating both sides with the condition that as time changes from 0 to t, the velocity changes from u to v. For the constant acceleration,

$$\int_u^v dv = \int_0^t a\,dt = a\int_0^t dt \Rightarrow [v]_u^v = a[t]_0^t$$

$$v - u = at \quad (or) \quad v = u + at \qquad \rightarrow (2.7)$$

Note If *a* is dependent on time *t*, *a* cannot be brought outside the time integral.

Displacement – time relation

(ii) The velocity of the body is given by the first derivative of the displacement with respect to time.

$$v = \frac{ds}{dt} \text{ or } ds = vdt$$

and since $v = u + at$,

We get $ds = (u + at)dt$

Assume that initially at time t = 0, the particle started from the origin. At a later time t, the particle displacement is s. Further assuming that acceleration is time-independent, we have

$$\int_0^s ds = \int_0^t u\,dt + \int_o^t at\,dt \;\; (or) \;\; s = ut + \frac{1}{2}at^2 \quad (2.8)$$

Velocity – displacement relation

(iii) The acceleration is given by the first derivative of velocity with respect to time.

$$a = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = \frac{dv}{ds}v$$

[since ds/dt = v] where s is displacement traversed.

This is rewritten as $a = \frac{1}{2}\frac{d}{ds}(v^2)$

$$\text{or } ds = \frac{1}{2a}d(v^2)$$

Integrating the above equation, using the fact when the velocity changes from u to v, displacement changes from 0 to s, we get

$$\int_0^s ds = \int_u^v \frac{1}{2a}d\left(v^2\right)$$

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